

find out if it exists or not. If it exists, it returns 'true', else 'false'.

Adding a book: When the button 'Add' is clicked, it invokes the procedure *isDataEntered*.

- If a 'false' is returned from the procedure (i.e., data was not entered in text fields), it displays an error message.
- If 'true' is returned, then we invoke the procedure called *isExistingBook*. The procedure checks to see if data already exists in our list. If it does, an error message is displayed; otherwise a join of title and author, in the format of *Title: Author*, is stored in the list. It also stores this latest data into our database using our TAG variable.

Viewing books in ListPicker:

When the 'View' button is pressed, we invoke the *ListPicker.Open* to show data. This will cause the *BeforePicking* block of *ListPicker* to be triggered. Once triggered, we set its elements to *listBooks*, which contains our book collection data. Next, the picker will open, showing the data.

Data in the picker (Figure 9) is displayed in the format 'BookTitle:BookAuthor'. When a selection is made, *AfterPicking* will be invoked. At this time, we take data for the current select, split it at the colon (:) and assign it to a temporary variable (*varTemp*) as a list. The list will have two items, with the first item as the *BookTitle* and the second item as *BookAuthor*. The next blocks take the first item (*BookTitle*) and set it into the text field (*txtBookTitle*), and then take the second item (*BookAuthor*) and set it into the text field (*txtBookAuthor*).

Deleting books: To delete a book, you can select an existing book from the picker (see Figure 10). You can also manually enter the title/author into the text fields and then click the 'Delete' button. Once clicked, we invoke

the procedure *isExistingBook*. The procedure (see above) checks to see if the book exists in the list and, based on that, returns 'true' or 'false'. If false (book not found), it displays an error

message. If found, we join the data from the text field using a colon (:). Next, we check for this *Title:Author* and get its index from our list variable *listBooks*. The result (index) is saved

into our temp var *varTemp*. Then, using that index number, we remove the item from our list. We then save the updated list of books into the database using the TAG we had defined as the key.

Resetting the database (book lists): To re-initialise the database and remove all books (reset the list), we have provided a button *btnClearList* (clear book list). The blocks in Figure 11 show how to reset the book list and re-init the database.

Now you are done with the block editor too. Next, let's move to download and install the app on your phone to check how it is working.

Packaging and testing

To test the app, you need to get it on your phone. First, download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the *Build* button. It will show you the option to download the apk to your computer.
2. While downloading, you can see its progress and after it has been successfully completed, the application will be placed in the *Download* folder of your directory or the preferred location you have set for it.
3. Now you need to get this apk file to your mobile phone either via Bluetooth or USB cable. Once you have placed the apk file on your SD card, you need to install it. Follow the on-screen instructions to do so. You might get some notification or warning saying that the install is from an untrusted source. Allow this from the settings, and after successful

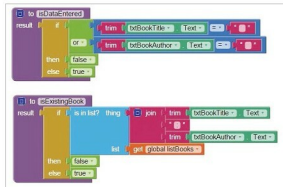


Figure 6: Block Editor Image 2

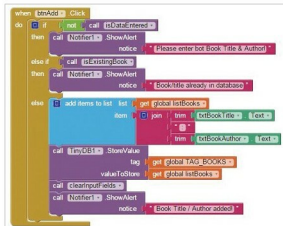


Figure 7: Block Editor Image 3

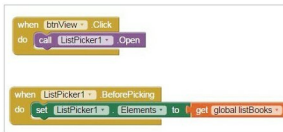


Figure 8: Block Editor Image 4

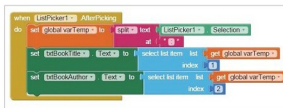


Figure 9: Block Editor Image 5